SOA Survey

Dependency-Aware Multi-Phase Liver Tumor Segmentation
with Explicit Phase Interaction Modeling

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Abstract

Liver tumor segmentation from contrast-enhanced computed tomography (CT) is a critical task in clinical oncology because accurate delineation of tumor boundaries supports diagnosis, treatment planning, and longitudinal monitoring. In clinical practice, multi-phase CT protocols acquire images at distinct temporal stages—non-contrast, arterial, portal venous, and delayed phases—each revealing complementary hemodynamic and tissue-contrast information. However, complete multi-phase acquisition is often unavailable due to scan time, radiation dose concerns, patient motion, or protocol variation across hospitals. This survey studies recent deep learning approaches for liver tumor segmentation with emphasis on multi-phase CT fusion, phase interaction modeling, and robustness to incomplete phase inputs. We reviewed both broad multimodal medical imaging surveys and task-specific liver segmentation papers closest to our direction, especially methods such as PA-Net, PA-ResSeg, MULLET, Diff4MMLiTS, and MCDA. Across the surveyed papers, a consistent pattern emerges: current methods improve segmentation through attention-based fusion, synthesis, or adaptive phase aggregation, but most learn phase relationships only implicitly. They do not explicitly model structured dependencies among phase combinations or higher-order phase interactions. Based on this gap, we propose to investigate explicit modeling of cross-phase dependencies for liver tumor segmentation with missing CT phases, with the goal of improving robustness, interpretability, and segmentation quality under realistic incomplete-input settings.

1 Problem Definition

Liver cancer is the sixth most common cancer worldwide and the fourth leading cause of cancer-related mortality. Hepatocellular carcinoma (HCC) accounts for 75–85% of primary liver tumors, and the liver is also one of the most frequent sites for metastatic disease from colorectal, breast, and lung cancers. Accurate segmentation of liver tumors is clinically essential because it supports diagnosis, surgical planning, treatment response assessment, and dosimetry calculation for interventional therapies such as transarterial radioembolization. Computed tomography (CT) is the principal imaging modality for liver tumor evaluation, valued for its high spatial resolution, fast acquisition, and standardized Hounsfield Unit (HU) measurement across scanners [12, 7].

In clinical liver imaging, contrast-enhanced CT (CECT) protocols acquire scans at multiple temporal phases after intravenous injection of iodinated contrast. The standard multi-phase protocol includes the non-contrast (NC) phase, the late hepatic arterial phase (AP, approximately 35–45 seconds post-injection), the portal venous phase (PVP, approximately 60–75 seconds), and the delayed or equilibrium phase (DP, 3–5 minutes). These phases are not redundant views of the same anatomy; each provides distinct hemodynamic information. The arterial phase highlights hypervascular lesions such as HCC, which receives blood predominantly from the hepatic artery. The portal venous phase provides the best overall liver-to-lesion contrast for hypovascular lesions such as metastases, and it reveals the characteristic washout pattern used for HCC diagnosis under the Liver Imaging Reporting and Data System (LI-RADS) [13]. The delayed phase captures equilibrium enhancement patterns useful for characterizing fibrotic components and cholangiocarcinoma. The non-contrast phase establishes baseline tissue attenuation before contrast administration [1, 2, 5]. Arterial hyperenhancement followed by portal venous or delayed washout is the defining imaging hallmark of HCC under current clinical guidelines [13], making multi-phase protocols a diagnostic necessity rather than a technical preference. As a result, liver tumor segmentation systems perform best when all phases are available together because tumor characterization depends on the temporal enhancement pattern across phases, not on any single scan alone.

Despite the diagnostic value of multi-phase imaging, this task remains difficult even in the complete-phase setting because liver tumors vary substantially in size, shape, number, and contrast with surrounding parenchyma. The LiTS benchmark demonstrates this heterogeneity across its 201 CT volumes from seven clinical centers, where lesion-to-background contrast varies from hyper- to hypo-dense and tumor count ranges from one to over ten per patient [12]. Tumor boundaries are often indistinct, especially in the presence of cirrhosis or steatosis, and segmentation quality is affected by respiratory motion artifacts, inter-phase spatial misalignment, and limited labeled data. The problem becomes harder when some CT phases are missing. In real clinical settings, complete multi-phase CT is not always available because of scan time constraints, radiation dose minimization, patient motion between acquisitions, contrast agent contraindications, or protocol variation across hospitals [8, 9]. From a deep learning perspective, the core challenge is not only incomplete input but the fact that existing methods mostly rely on implicit feature fusion, attention-based aggregation, or synthesis without explicitly modeling how phase combinations contribute to tumor characterization. Higher-order relationships—such as how the interaction between arterial, portal venous, and delayed phases jointly characterizes HCC versus metastasis—remain weakly understood. This makes the problem simultaneously one of representation learning, multi-phase fusion, and structured dependency modeling under incomplete data.

2 Literature Review

2.1 From Multi-Phase CT Imaging to Liver Tumor Segmentation

Medical imaging for liver cancer diagnosis is inherently multi-phase. Surveys on deep learning for multimodal liver cancer segmentation describe clinical liver imaging as a combination of complementary temporal acquisitions, where different CT phases each reveal distinct tissue properties [1, 2]. The multimodal machine learning literature formalizes the broader challenge of learning from heterogeneous data through the lens of representation, fusion, alignment, translation, and co-learning [3]. This is useful conceptual framing: it situates multi-phase CT within the wider multimodal ML landscape and establishes that implicit fusion is insufficient across medical imaging domains generally. However, [3] is a broad biomedicine survey and does not directly establish the specific gap around explicit multi-phase liver dependency modeling; it is used here for framing only, not for task-specific liver claims.

Liver tumor segmentation is therefore one of the clearest examples of a multi-phase problem, since tumor characterization depends not on one scan alone but on how several CT phases complement each other [1, 2].

2.2 Why CT Phases Matter and Why Missing Ones Create a Hard Problem

In the liver tumor setting, the literature repeatedly focuses on the standard CECT phases as clinically distinct temporal views rather than interchangeable inputs. The arterial phase mainly highlights hypervascular lesions and active tumor regions receiving hepatic arterial blood supply; this is clinically critical because HCC can be diagnosed by imaging due to its characteristic vascular patterns [1, 13]. The portal venous phase provides the highest overall liver parenchymal enhancement and is most sensitive for detecting hypovascular metastases, which is why most single-phase liver segmentation studies default to this phase [6, 7]. The delayed phase captures late enhancement patterns characteristic of cholangiocarcinoma and fibrotic components. The non-contrast phase establishes baseline tissue density [1, 2, 5].

Because each phase contributes different hemodynamic evidence, segmentation and diagnostic performance depend strongly on using them together. PA-ResSeg demonstrated that incorporating arterial phase features alongside portal venous phase features via inter-phase attention improved Dice per case from the single-phase baseline by a meaningful margin [7]. PA-Net showed a 3.3% mean Dice improvement over the single-phase nnUNet baseline when fusing arterial and portal venous phases [6]. However, this dependence on multiple phases also creates the field's most persistent bottleneck: complete multi-phase CT is often unavailable due to scan time, radiation dose, patient motion, contrast allergies, or protocol variation [8, 9]. Spatial misalignment between phases caused by respiratory motion is a major source of feature interference—PA-Net specifically addresses this through B-spline nonrigid registration [6]. Missing even one clinically important phase, especially the arterial or portal venous phase, can cause substantial performance degradation [6, 9].

What this group establishes: The clinical necessity of multi-phase CT and the measurable cost of missing phases. **What it does not solve:** Why specific phase combinations matter for specific tumor types, and how to model those relationships explicitly.

2.3 Challenge 1: Learning Under Missing or Incomplete Phases

The missing-phase setting is consistently identified as one of the most important practical challenges in multi-phase liver tumor analysis [4, 8, 9]. The systematic survey on missing modalities in medical imaging groups current methods into three dominant families: image synthesis, knowledge transfer, and latent-space methods [4]. In the liver-specific literature, these three strategies each have concrete representatives.

Diff4MMLiTS represents the image synthesis approach [8]. It uses a four-stage pipeline: (1) pre-registration of the target liver organ in multimodal CTs, (2) mask dilation and inpainting to obtain normal CTs without tumors, (3) synthesis of strictly aligned multi-phase CTs with tumors using a latent diffusion model conditioned on multi-phase CT features and random tumor masks, and (4) training the segmentation model on both real and synthetic data. This effectively transforms the missing-phase problem into a complete-phase one. However, synthesized scans may be over-smoothed, may hallucinate structures, and can propagate errors into the downstream segmenter [4, 8].

The H-LSTM framework represents a sequence-based adaptive approach [9]. It uses a shared image feature extractor across phases, an internal LSTM for each phase to aggregate features from different 2D CT layers, and an external LSTM across phases. This allows it to handle incomplete phases and varying numbers of CT image layers naturally. However, phase information is aggregated through recurrent hidden states without explicitly modeling which phase combinations matter for specific tumor types.

PA-Net addresses missing phases through a training strategy that preserves effective use of arterial phase information even when this phase is unavailable during inference, designing the pipeline so that the model learns robust representations from whichever phases are present [6]. Across all three families, the same criticism appears: these methods improve robustness but do not explicitly model how phases depend on one another or provide interpretable phase contribution scores [1, 4].

What this group solves: Practical robustness to missing phases through synthesis, recurrence, and training-time strategies.

What it still does not solve: Structured, interpretable modeling of which phase combinations matter and why—leaving the dependency structure itself unlearned.

2.4 Challenge 2: Fusion and Phase Interaction Modeling

The multimodal segmentation literature classifies fusion strategies into input-level (simple concatenation), layer-level (interaction inside the network), and decision-level (late combination). Input-level fusion concatenates phases without reasoning about their relationships. Decision-level fusion allows independent processing but interaction occurs too late. Layer-level fusion is theoretically stronger but still typically learned implicitly [3].

PA-ResSeg introduced the Phase Attention (PA) block: Intra-PA uses squeeze-and-excitation style channel attention to capture within-phase dependencies, and Inter-PA models cross-phase interdependencies by learning how arterial phase features recalibrate portal venous phase features [7]. PA-Net extended this to voxel-level attention weight maps, producing per-voxel fusion decisions that better handle local variations in phase alignment [6]. MCDA proposed a multi-phase channel-stacked dual attention module that receives input from an arbitrary number of phases and dynamically assigns weights, combined with a scale-weighted loss for multi-scale supervision [10].

MULLET introduces transformer-based attention for embedding multi-phase CECT contexts using ROI-guided feature embedding and cross-phase attention to handle misalignment, demonstrating strong detection performance on a 1,229-scan dataset [5].

These methods represent a clear progression from naive concatenation to richer interaction-aware designs. However, improved fusion does not guarantee genuine cross-phase understanding [1, 3]. PA-ResSeg's Inter-PA learns a recalibration function but does not output an interpretable dependency score. MCDA dynamically weights phases but does not formalize higher-order interactions such as how AP + PVP + DP combinations specifically aid HCC characterization.

What this group solves: Richer feature-level interactions through channel, voxel, and transformer-based fusion mechanisms.

What it still does not solve: Explicit, structured dependency rules across phase subsets, or interpretable outputs that explain phase contribution at the tumor-type level.

2.5 Challenge 3: Task-Specific Core Segmentation Methods

The strongest task-specific papers closest to our direction are PA-ResSeg, PA-Net, MULLET, Diff4MMLiTS, MCDA, and the multi-phase spatial aggregation method [5, 6, 7, 8, 10, 11]. These form the central comparison anchor because they directly address liver tumor segmentation using multi-phase CT.

PA-ResSeg is the foundational multi-phase baseline [7]. It achieves DPC of 0.7787 and DG of 0.8682 on MPCT-FLLs and DPC of 0.8290 on a second multi-phase dataset. Its 3D boundary-enhanced (BE) loss addresses the persistent boundary segmentation difficulty. PA-Net extends attention to voxel-level granularity and introduces a learning strategy for thick-layer CT images with inter-phase misalignment, achieving state-of-the-art results on a 362-patient in-house dataset [6]. MULLET proposes a transformer-based system evaluated on 1,229 CECT scans from 1,197 patients, achieving DPC of 78.47% [5]; it shows that transformer-based context embedding can achieve strong performance, though phase interactions remain implicit inside transformer attention. The multi-phase spatial aggregation method introduces a Spatial Aggregation Module (SAM) for per-pixel cross-phase interactions and an Uncertain Region Inpainting Module (URIM) to refine ambiguous boundary pixels [11]. MCDA proposes an insertable dual attention module evaluated on all four standard phases [10].

None of these methods models structured dependencies across phase combinations directly. PA-ResSeg and PA-Net learn recalibration weights but do not formalize which phase combinations matter for edema versus active tumor versus fibrosis [6, 7]. MCDA dynamically weights phases but does not learn higher-order interaction rules [10]. These papers show a clear progression from single-phase to richer interaction-aware methods, but stop short of making dependency patterns themselves the object of learning.

What this group solves: State-of-the-art liver tumor segmentation under multi-phase settings, with strong baselines for comparison.
What it still does not solve: Explicit, structured modeling of cross-phase dependencies with interpretable, clinically meaningful phase contribution scores.

2.6 Challenge 4: Representation Learning and Phase Selection

The representation-learning literature distinguishes between joint representations (modalities combined into one shared vector) and coordinated representations (separate embeddings constrained for cross-modal similarity) [3]. Liver segmentation has historically leaned toward joint fusion, while recent work increasingly recognizes the need to preserve phase-specific meaning while enforcing cross-phase consistency [1, 3].

H-LSTM uses hierarchical recurrent modeling to implicitly learn sequential phase representations [9]. Yet the representation-learning perspective makes the current limitation clearer: attention-guided or recurrence-based representations may align phases without ever learning an explicit rule for why a particular phase interaction matters for HCC versus metastasis versus cholangiocarcinoma [1].

Open Problems (flagged by the field):

- Most current methods still learn phase relationships implicitly, while the surveyed literature collectively suggests the need for more structured and interpretable cross-phase modeling [1, 4].
- Existing methods remain weak in interpretability and often conflate improved fusion with genuine understanding of phase interactions [1, 3].
- Higher-order combination learning—modeling how sets like AP + PVP + DP jointly contribute to segmentation of specific tumor types—remains insufficiently explored [6, 7, 10].

What this group solves: Adaptive phase selection and sequence-based handling that begin to differentiate phase importance.
What it still does not solve: The jump from feature-level phase importance to segmentation-level structured dependency rules with interpretable outputs aligned to clinical reasoning.

3 Research Gap and Proposed Direction

The literature shows that multi-phase liver tumor segmentation under missing or incomplete CT phases is an increasingly active and clinically relevant problem setting, but existing methods largely attack it through attention-based fusion, synthesis, recurrent aggregation, or adaptive phase weighting. Surveys and core task-specific papers agree that these strategies improve segmentation quality, yet they still learn phase relationships only implicitly. Fusion-based methods such as PA-Net and PA-ResSeg combine phase information through attention weights without clearly identifying how phase combinations contribute to tumor characterization. Synthesis methods like Diff4MMLiTS reconstruct missing phases but do not formalize structured dependency patterns. Even the closest interaction-aware methods like MCDA and the spatial aggregation method remain mostly feature-level and do not explicitly learn higher-order relationships among subsets such as AP + PVP + DP.

Although prior work does not always frame this limitation explicitly as dependency learning, the surveyed evidence collectively points to the absence of structured, interpretable modeling of phase combinations. The gap we identify is therefore a synthesis of converging evidence across the literature rather than a phrase used identically by prior work.

Among the possible gaps in this area, we focus on explicit modeling of cross-phase dependencies. This gap is significant because it sits at the center of both robustness and interpretability: if a model can learn how phases depend on one another, it may better handle missing inputs while also providing a more principled understanding of which phase combinations matter for different tumor types. It is also tractable within the project scope because the literature already provides strong baselines but leaves room for a new dependency-oriented formulation. Specifically, we envision a module that can produce quantifiable outputs such as per-phase importance scores (e.g., "arterial phase contributed X%") and per-combination interaction scores (e.g., "AP + PVP combination was most critical for this HCC segmentation"), aligning model reasoning with clinical diagnostic patterns.

Summary Gap Statement: Prior work improves liver tumor segmentation under multi-phase CT settings through attention-based fusion, synthesis, and adaptive interaction, but struggles to explicitly model structured dependencies among phase combinations; we therefore propose to investigate explicit cross-phase dependency learning for robust liver tumor segmentation with incomplete CT inputs.

Research Question: How can explicit modeling of cross-phase dependency patterns improve liver tumor segmentation when one or more CT phases are missing? In particular, can structured dependency learning provide better robustness and interpretability than existing implicit fusion-based methods, and can it produce clinically meaningful phase contribution scores aligned with radiological reasoning?

4 Benchmark, Evaluation, and Data Landscape

Datasets. Datasets relevant to this survey fall into two groups: core multi-phase CT datasets most directly relevant to the problem setting, and supporting context datasets that provide supplementary evidence.

Core datasets. The LiTS (Liver Tumor Segmentation Benchmark) dataset comprises 201 contrast-enhanced abdominal CT volumes (131 training, 70 test) from seven clinical institutions, with pixel-wise annotations for liver and intrahepatic tumors including both primary and metastatic lesions [12]. The MPCT-FLLs (Multi-Phase CT dataset of Focal Liver Lesions) is an in-house multi-phase benchmark used by PA-ResSeg containing four-phase CT scans (NC, AP, PVP, DP) with annotated focal liver lesions; it is not publicly available [7]. The AIELDI dataset used by MULLET contains 1,229 multi-phase CECT scans from 1,197 patients with expert-annotated lesions [5].

A practical limitation of this research direction is that the strongest truly multi-phase CT liver lesion datasets used by recent methods are private, while LiTS remains the main public benchmark but is not a dedicated public missing-phase multi-phase CT benchmark, limiting reproducibility under realistic multi-phase settings.

Supporting context datasets. 3D-IRCADb is a small CT dataset with liver and tumor annotations useful as a supporting reference [2]. CHAOS is a multi-organ abdominal benchmark covering CT and MRI, useful as supporting context but not a direct multi-phase liver tumor benchmark [2].

Table 1 summarizes all datasets with the core/supporting distinction made explicit.

Evaluation Protocol. The most important evaluation metric in this field is Dice Similarity Coefficient (DSC) because segmentation is fundamentally an overlap problem between predicted tumor masks and expert annotations. The LiTS benchmark uses both Dice per case (DPC) and global Dice (DG); PA-ResSeg reports both metrics along with Volumetric Overlap Error (VOE) and Relative Volume Difference (RVD) [7, 12]. MULLET additionally reports precision, recall, and F2 score to balance detection and segmentation quality [5]. Some works also report HD95 for boundary quality. For our problem, DSC/DPC is the primary metric as it is the standard in LiTS-style evaluation, while VOE, HD95, and detection metrics serve as useful secondary measures.

Compute Context. PA-ResSeg reports training on NVIDIA GPUs with 3D multi-phase input volumes using a ResNet-based backbone with multi-scale PA blocks [7]. MULLET uses transformer-based architectures trained on 1,229 CECT volumes [5]. Diff4MMLiTS employs a latent diffusion model requiring substantial GPU memory for both synthesis and segmentation stages [8]. Several 3D transformer- and diffusion-based multi-phase pipelines are computationally demanding and commonly benefit from high-memory V100/A100-class hardware; this should be factored into experimental planning accordingly.

Experimental Design Considerations. A rigorous experimental design should compare the proposed dependency-oriented approach against the strongest available multi-phase baselines: PA-Net and PA-ResSeg (attention-based fusion), MCDA (adaptive multi-phase dual attention), and the spatial aggregation method (per-pixel phase interaction). Additional comparisons with Diff4MMLiTS (synthesis-based) and H-LSTM (sequence-based) would cover alternative missing-phase strategies.

Meaningful ablations should isolate whether improvement comes from explicit dependency modeling versus increased model capacity. The design should compare: (1) a fusion-only baseline using naive phase concatenation, (2) a model replicating PA-Net-style channel/voxel attention, (3) a model with pairwise dependency modeling only, and (4) a model with higher-order dependency modeling across phase subsets (e.g., all $2^4 - 1 = 15$ non-empty subsets of four phases). Another ablation should compare full-phase versus controlled missing-phase settings, especially clinically important cases where AP or PVP is absent. An interpretability ablation should compare learned phase importance scores against known clinical priors (e.g., AP should score highly for HCC, PVP for metastases).

Several evaluation pitfalls should be considered. First, simulated missing phases via random masking may not reflect real hospital settings where specific phases are absent due to protocol-level decisions; missing-phase experiments should model plausible clinical acquisition patterns. Second, relying only on Dice can hide boundary errors—VOE and HD95 should accompany DPC. Third,

Table 1: Key datasets relevant to multi-phase liver tumor segmentation. Core datasets are directly relevant to the problem; supporting datasets provide supplementary context.

Dataset	Size	Phases	Access	Notes
Core datasets for multi-phase liver CT segmentation				
LiTS [12]	201 CTs (131/70)	CE-CT (PVP)	Public	Primary public benchmark; multi-center, 7 institutions.
MPCT-FLLs [7]	In-house	NC, AP, PVP, DP	Private	Used by PA-ResSeg; not publicly available.
AIELDI [5]	1,229 CTs (1,197 pts)	Multi-phase CECT	Private	Used by MULLET; expert-annotated.
Supporting context datasets				
3D-IRCADb [2]	20 CTs	CE-CT	Public	Small CT dataset; not multi-phase.
CHAOS [2]	40 CT + 40 MRI	CT/MRI	Public	Multi-modal abdominal benchmark; supporting context only.

the LiTS benchmark's multi-center heterogeneity demonstrates that cross-scanner variation affects generalization. Fourth, inter-phase misalignment from respiratory motion must be handled during preprocessing (e.g., B-spline registration as in PA-Net). Finally, a meaningful validation should include not only overlap performance but also robustness across missing-phase patterns and qualitative evidence that the model produces clinically plausible phase dependency scores.

5 Annotated Reading List

References

[1] C. Wu et al., "A Review of Deep Learning Approaches for Multimodal Image Segmentation of Liver Cancer," *Journal of Applied Clinical Medical Physics*, vol. 25, no. 12, p. e14540, 2024.
What it does: Reviews deep learning techniques for multimodal fusion image segmentation of liver cancer, covering architectures from CNN and U-Net through Transformer, GAN, and SAM.
What it contributes: Provides a comprehensive taxonomy of fusion methods (input, feature, decision-level) in the liver cancer setting and identifies multi-phase fusion as an open research direction.
Why it is here: Foundational survey for the landscape of liver cancer segmentation; used for broad framing of multi-phase fusion challenges and phase importance claims, not as direct evidence for specific dependency modeling gaps.

[2] S. Gul, M. S. Khan, A. Bibi, A. Khandakar, M. A. Ayari, and M. E. H. Chowdhury, "Deep Learning Techniques for Liver and Liver Tumor Segmentation: A Review," *Computers in Biology and Medicine*, vol. 147, p. 105620, 2022.
What it does: Surveys deep learning architectures for CT and MR liver and liver tumor segmentation, tracing the evolution from classical CNNs to advanced encoder-decoder designs.
What it contributes: Establishes the shift from handcrafted methods to deep learning and highlights challenges including tumor variability, limited labeled data, and the importance of multi-phase information.
Why it is here: Useful background on liver segmentation challenges and for positioning multi-phase methods within the broader architectural landscape.

[3] E. Warner, J. Lee, W. Hsu, T. Syeda-Mahmood, C. E. Kahn Jr., O. Gevaert, and A. Rao, "Multimodal Machine Learning in Image-Based and Clinical Biomedicine: Survey and Prospects," *International Journal of Computer Vision*, vol. 132, no. 9, pp. 3753–3769, 2024.
What it does: Defines multimodal learning through five core challenges: representation, fusion, alignment, translation, and co-learning.
What it contributes: Distinguishes joint from coordinated representations and highlights missing modalities and cross-modal relationships as unresolved real-world issues.
Why it is here: Positions multi-phase liver imaging within the wider multimodal ML landscape; supports the argument that implicit fusion is insufficient across medical imaging domains.

[4] J. Qian, A. Joshi, H. Li, N. A. Parikh, J. R. Dillman, and L. He, "Utility of Deep Learning to Address Missing Modalities from Multi-Modal Medical Imaging Studies: A Systematic Review," *Artificial Intelligence and Applications*, 2025.
What it does: Organizes missing-modality learning into image synthesis, knowledge transfer, and latent-space methods with a systematic analysis of 78 studies.
What it contributes: Shows that current methods reconstruct, transfer, or fuse information without explicit dependency learning and flags this as a persistent limitation across the field.
Why it is here: The strongest cross-domain survey supporting the claim that structured modality depen-

dency modeling is still absent across medical imaging.

[5] L. Wu, H. Wang, Y. Chen, X. Zhang, T. Zhang, N. Shen, G. Tao, Z. Sun, Y. Ding, W. Wang, and J. Bu, "Beyond Radiologist-Level Liver Lesion Detection on Multi-Phase Contrast-Enhanced CT Images by Deep Learning," *iScience*, vol. 26, no. 11, p. 108183, 2023.
What it does: Proposes MULLET, a transformer-based system for multi-phase liver lesion detection and segmentation evaluated on 1,229 CECT scans from 1,197 patients, achieving DPC of 78.47% and outperforming two expert radiologists.
What it contributes: Demonstrates that transformer-based multi-phase context embedding with ROI-guided attention can achieve strong performance on misaligned multi-phase CECT data.
Why it is here: Core paper showing multi-phase learning capabilities; phase interactions remain implicit inside transformer attention and no explicit dependency scores are produced.

[6] Z. Liu, J. Hou, X. Pan, R. Zhang, and Z. Shi, "PA-Net: A Phase Attention Network Fusing Venous and Arterial Phase Features of CT Images for Liver Tumor Segmentation," *Computer Methods and Programs in Biomedicine*, vol. 244, p. 107997, 2024.
What it does: Proposes voxel-level phase attention for fusing arterial and portal venous phase CT features, plus a learning strategy for handling inter-phase misalignment via B-spline registration; evaluated on a 362-patient in-house dataset.
What it contributes: Achieves 3.3% mean Dice improvement over single-phase nnUNet; provides a missing-phase handling scheme through its training strategy.
Why it is here: One of the closest papers to our direction; foregrounds phase interaction, but attention remains feature-level without formalized dependency scores or higher-order combination rules.

[7] Y. Xu, M. Cai, L. Lin, Y. Zhang, H. Hu, Z. Peng, Q. Zhang, Q. Chen, X. Mao, Y. Iwamoto, and X. Han, "PA-ResSeg: A Phase Attention Residual Network for Liver Tumor Segmentation from Multiphase CT Images," *Medical Physics*, vol. 48, no. 7, pp. 3752–3766, 2021.
What it does: Introduces Intra-PA (channel-wise self-dependencies) and Inter-PA (cross-phase interdependencies) modules within a multi-scale fusion architecture, plus a 3D boundary-enhanced loss.
What it contributes: Achieves DPC of 0.7787 and DG of 0.8682 on MPCT-FLLs; demonstrates generalization to a second dataset (DPC 0.8290) and across different backbones.
Why it is here: Foundational multi-phase segmentation baseline; its Inter-PA module is the earliest explicit inter-phase attention mechanism in liver segmentation, but still learns dependencies implicitly.

[8] S. Chen, S. Dou, L. Pan, H. Jin, Y. Zhang, Z. He, Y. Zheng, and H. Wang, "Diff4MMLiTS: Advanced Multimodal Liver Tumor Segmentation via Diffusion-Based Image Synthesis and Alignment," in *MLMI (MICCAI Workshop)*, LNCS vol. 16241, 2025.
What it does: Uses a four-stage pipeline—registration, inpainting via Normal CT Generator, synthesis via Multimodal CT Synthesizer with latent diffusion, and multimodal segmentation—to generate aligned multi-phase CTs.
What it contributes: Translates the tumor registration challenge into a data synthesis problem, eliminating the need for perfectly registered multi-phase data.
Why it is here: Represents the synthesis approach to missing-phase handling; does not model phase dependencies or provide interpretable phase contribution analysis.

[9] S. Huang, X. Nie, K. Pu, Y. Li, L. Shen, L. Li, J. Yuan, and C. Li, "A Flexible Deep Learning Framework for Liver Tumor Diagnosis Using Variable Multi-Phase Contrast-Enhanced CT Scans," *Journal of Cancer Research and Clinical Oncology*, vol. 150, p. 443, 2024.
What it does: Designs a Hierarchical LSTM with shared feature extractors, internal LSTMs per phase, and external LSTMs across phases, tested on 276 participants (94 HCC, 99 ICC, 83 normal).
What it contributes: Handles incomplete phases and varying CT layer counts naturally through recurrent aggregation, making it suitable for real-world variable-protocol scenarios.
Why it is here: Sequence-based approach to multi-phase handling; its recurrent strategy contrasts with our proposed explicit dependency formulation.

[10] H. Kuang, X. Yang, H. Li, J. Wei, and L. Zhang, "Adaptive Multi-Phase Liver Tumor Segmentation with Multi-Scale Supervision," *IEEE Signal Processing Letters*, vol. 31, pp. 426–430, 2024.
What it does: Proposes a Multi-phase Channel-stacked Dual Attention (MCDA) module that receives input from an arbitrary number of phases, dynamically assigns weights, and uses a scale-weighted loss for multi-scale supervision.
What it contributes: Demonstrates that dynamic phase weighting with channel stacking outperforms fixed fusion strategies and that multi-scale supervision reduces over-segmentation.
Why it is here: Closest existing work to adaptive

phase weighting for liver segmentation; its attention mechanism captures phase importance implicitly without structured higher-order dependency modeling.

[11] Y. Zhang, F. Peng, Y. Xu, H. Hu, and Y. Yu, "Multi-phase Liver Tumor Segmentation with Spatial Aggregation and Uncertain Region Inpainting," in *MICCAI*, LNCS vol. 12901, pp. 68–77, 2021.
What it does: Introduces a Spatial Aggregation Module (SAM) for per-pixel cross-phase interactions and an Uncertain Region Inpainting Module (URIM) to refine ambiguous boundary pixels using neighboring features.
What it contributes: Moves beyond channel-level fusion to per-pixel cross-phase interaction and addresses segmentation uncertainty explicitly at the boundary level.
Why it is here: Important for understanding the progression from channel attention to spatial-level phase interaction in liver segmentation.

[12] P. Bilic, P. F. Christ, H. B. Li, E. Vorontsov et al., "The Liver Tumor Segmentation Benchmark (LiTS)," *Medical Image Analysis*, vol. 84, p. 102680, 2023.
What it does: Reports the LiTS benchmark with 201 CT volumes from seven institutions, 75 submitted algorithms, and standardized evaluation across ISBI 2017, MICCAI 2017, and MICCAI 2018.
What it contributes: Establishes the dominant public benchmark; best tumor Dice per case of 0.739 at MICCAI 2018; reveals that top segmentation methods do not necessarily excel at tumor detection.
Why it is here: Essential benchmark dataset; its multi-center heterogeneity makes it ideal for evaluating robustness of dependency-aware approaches.

[13] American College of Radiology, "Liver Imaging Reporting and Data System (LI-RADS) v2018," American College of Radiology, Reston, VA, 2018. [Online]. Available: https://www.acr.org/Clinical-Resources/Reporting-and-Data-Systems/LI-RADS
What it does: Defines the standardized criteria for interpreting liver CT and MRI findings, including the multi-phase enhancement patterns used to characterize and diagnose HCC and other lesions.
What it contributes: Establishes the clinical basis for why specific CT phases matter: arterial hyperenhancement and portal venous/delayed washout are the defining radiological criteria for HCC diagnosis.
Why it is here: Direct clinical reference for LI-RADS scoring, HCC vascularity, and washout patterns discussed in the problem definition; grounds the phase importance claims in authoritative clinical criteria.

6 Preliminary Implementation Resources

- https://github.com/Houjunfeng203934/PA-Net
  PA-Net — official implementation reference. Provides the voxel-level phase attention fusion pipeline for multi-phase liver tumor segmentation, including data preprocessing, B-spline registration, training scripts, and evaluation code. Directly useful as a multi-phase fusion baseline.

- https://github.com/csying/PA-ResSeg
  PA-ResSeg — official implementation. Implements the intra-phase and inter-phase attention modules within the multi-scale fusion architecture, plus the 3D boundary-enhanced loss. Key baseline for multi-phase liver tumor segmentation and for reusing the PA block design.

- https://github.com/yzhang-zju/multi_phase_LiTS
  Multi-phase LiTS Spatial Aggregation — official implementation. Implements the SAM and URIM modules for per-pixel cross-phase interaction and uncertain region inpainting. Directly relevant as a spatial-level fusion baseline.

- https://github.com/ydk122024/MCDA
  MCDA — official implementation. Provides the multi-phase channel-stacked dual attention module with scale-weighted loss. Relevant as the most recent adaptive multi-phase segmentation baseline.

- https://github.com/MIC-DKFZ/nnUNet
  nnU-Net. The self-configuring medical image segmentation framework achieving state-of-the-art results across numerous benchmarks including LiTS. Useful as a strong automated single-phase baseline and for reusing its data preprocessing pipeline.